TOURIST RENTALS AND HOUSING MARKETS:

A COUNTERFACTUAL ANALYSIS OF TOURISTIC FLATS LICENSE CANCELLATION IN BARCELONA

Ana Cláudia da Costa

WHAT HAPPENS TO HOUSING PRICES IF ALL TOURIST FLATS LICENSES ARE CANCELLED?

Hypothesis

- Removing Touristic Flats reduce selling prices?
- Removing Touristic Flats reduce rental prices?

MOTIVATION

Housing affordability is a major public concern, and Barcelona faces an intense tourism pressure.

The city government announced that by 2028 all the hut licences will not be renewed.

What will be the impact on the housing market?

Let’s try to answer it.

GLOSSARY – KEY TERMS

- **HUT** – Tourist Flats (Habitatge d'Ús Turistic)
- **HUT Density** – Number of HUTs per hectare
- **Baseline Prediction** – Predicted price with existing HUTs
- **Counterfactual Prediction** – Predicted price assuming no HUTs
- **Impact (%)** – Percentage difference between baseline and counterfactual

IMPORTANT NOTE:

"This is **not a forecast** of what will actually happen, but a **model-based simulation** of what could happen if HUTs disappear, assuming that the relationships observed today remain stable.

NEWS!!!

idealista/news

Buscar noticias, reportajes

Suscribirse al boletín informativo

Iniciar sesión

Inmobiliario | Finanzas | Fiscalidad | Vacacional | Deco | Art | Especiales | Opinión | Foro | Estadísticas | Buscador de inmuebles

Vivienda | Oficinas | Retail | Empresas | Construcción | Top idealista | Internacional | Marketing inmobiliario | Blog de idealista | Hoteles

Vivienda
Las donaciones de padres a hijos para vivienda baten récords: así tributan en cada CCAA

Blog de idealista
idealista lanza su app en ChatGPT

Vivienda
Los países más poderosos del mundo por su 'poder blando'

Destinos turísticos
Los 6 miradores en Guadalajara para conocer sus pueblos y paisajes

Apartur carga contra el fin de los pisos turísticos en Barcelona: "No mejorará la vivienda"

La asociación asegura en Europa que la eliminación de los pisos turísticos en 2028 no aumentará la oferta residencial y causará un grave perjuicio económico y social

velocidad con la que se ha producido el cambio. Mientras tanto, el sector turístico clave en la economía local intenta adaptarse sin caer en el colapso reputacional.

Más Barcelona

L'Hospitalet Badalona

PLA VIURE

El Tribunal Co
en Barcelona

En 2028, el Ajunta
(VUT) para uso res
emergencia habita

n
nsa

ntud-- de nuestro

as a manos de
e las grandes
xtranjeros
rismo
so, es el
nuestro país,
problema real
a
estos días
e New York

DATA & TECH

python

pandas

python

scikit learn

machine learning in Python

GeoPandas

seaborn

statsmodels

matplotlib

DATASET OVERVIEW

73 Neighborhoods

Variables:

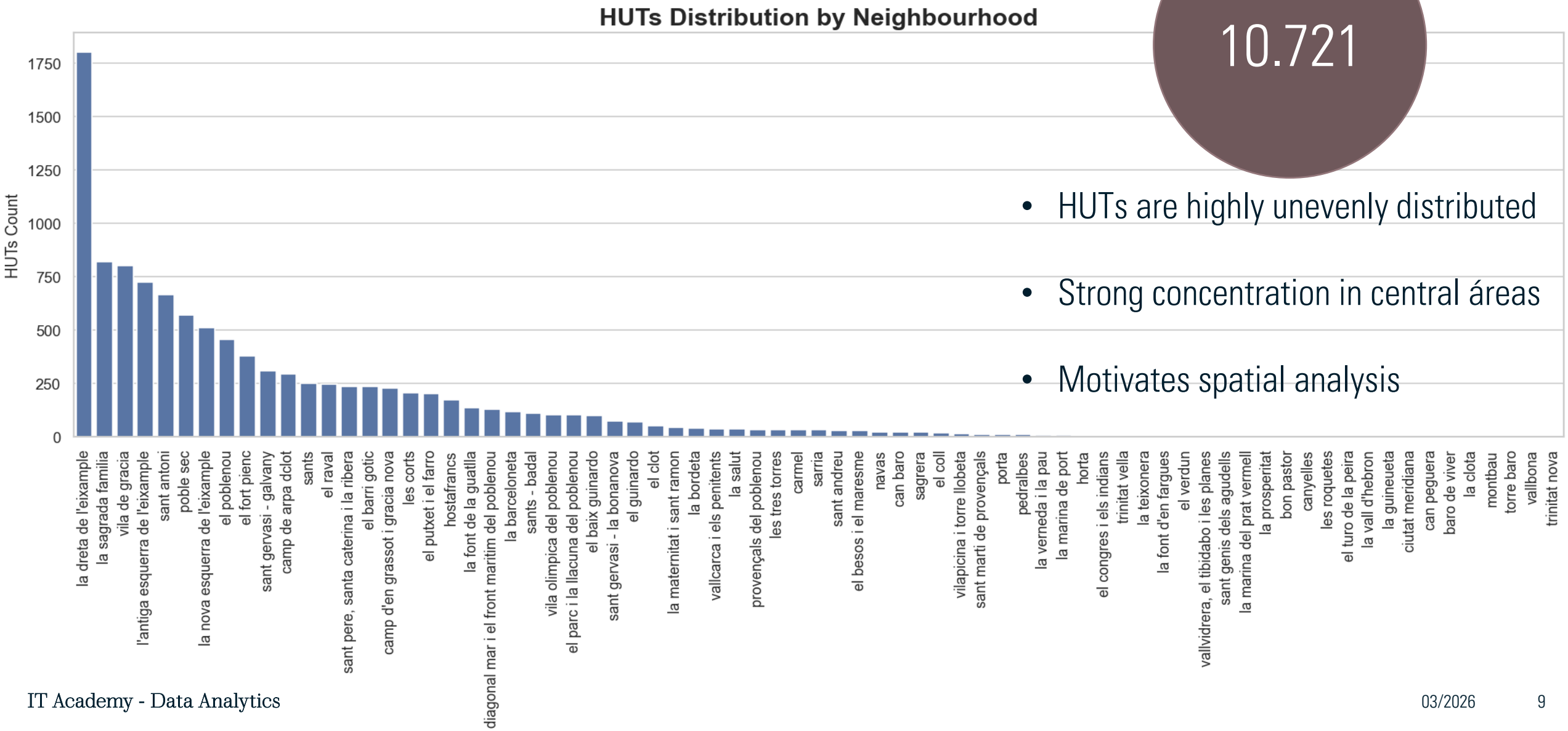
- Selling Price
- Rental Price
- Income per capita
- Population density
- Foreigh population density
- Hotel density
- HUT count
- HUT density

Barcelona

Open Data BCN

Barcelona's City Hall Open Data Service

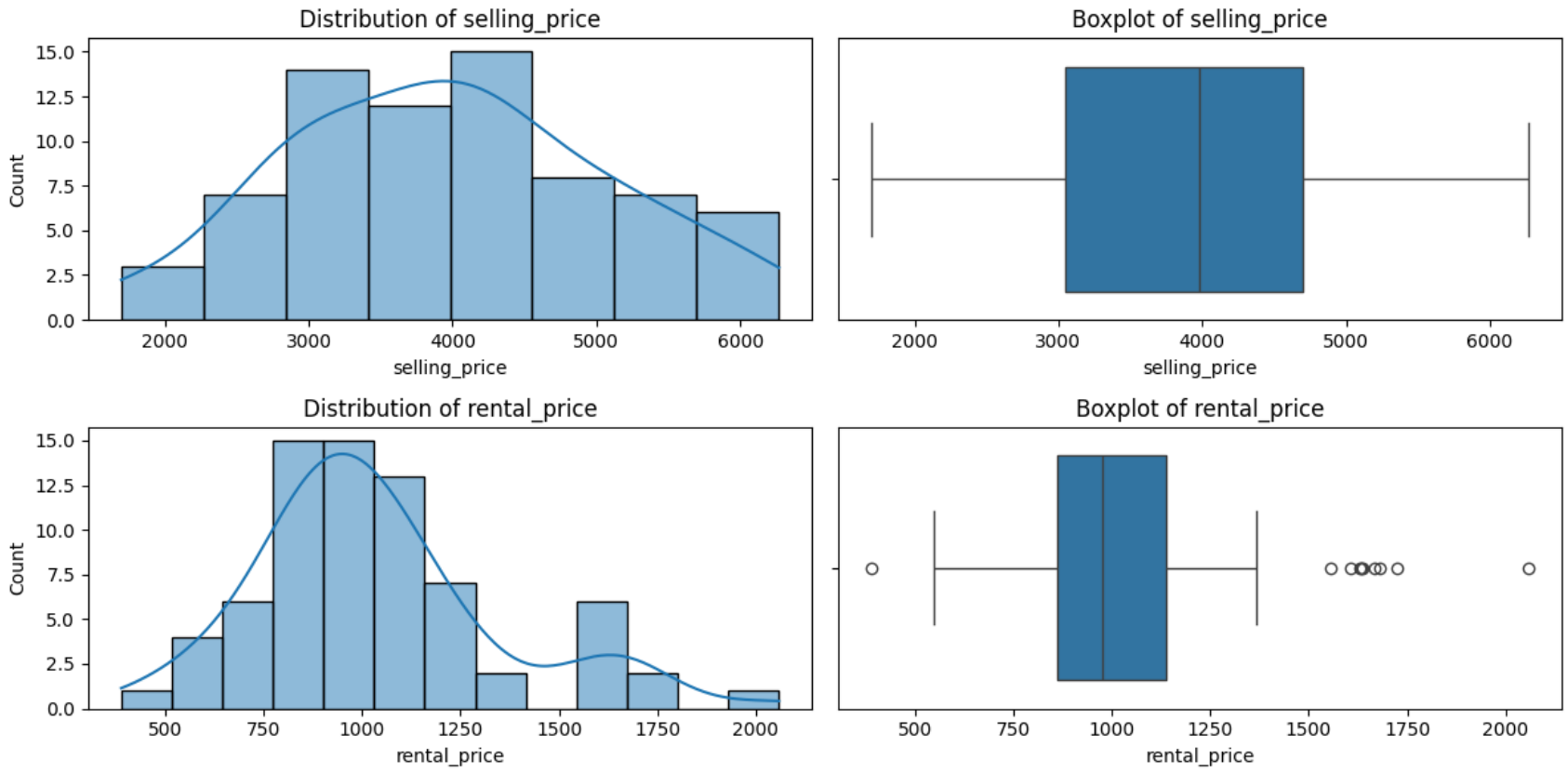
HUTS DISTRIBUTION



10.721

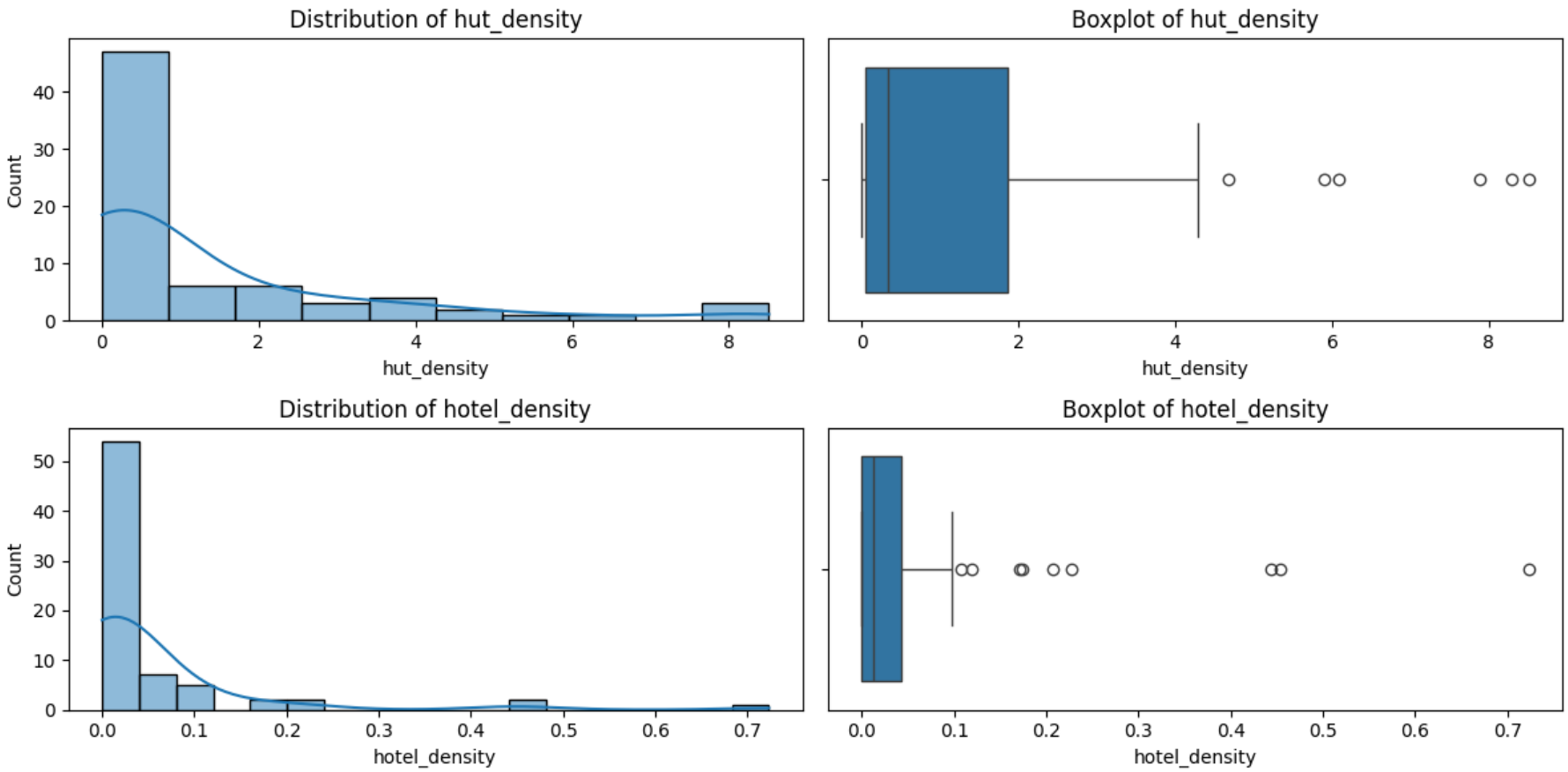
- HUTs are highly unevenly distributed
- Strong concentration in central áreas
- Motivates spatial analysis

VARIABLES DISTRIBUTION – HOUSING MARKET



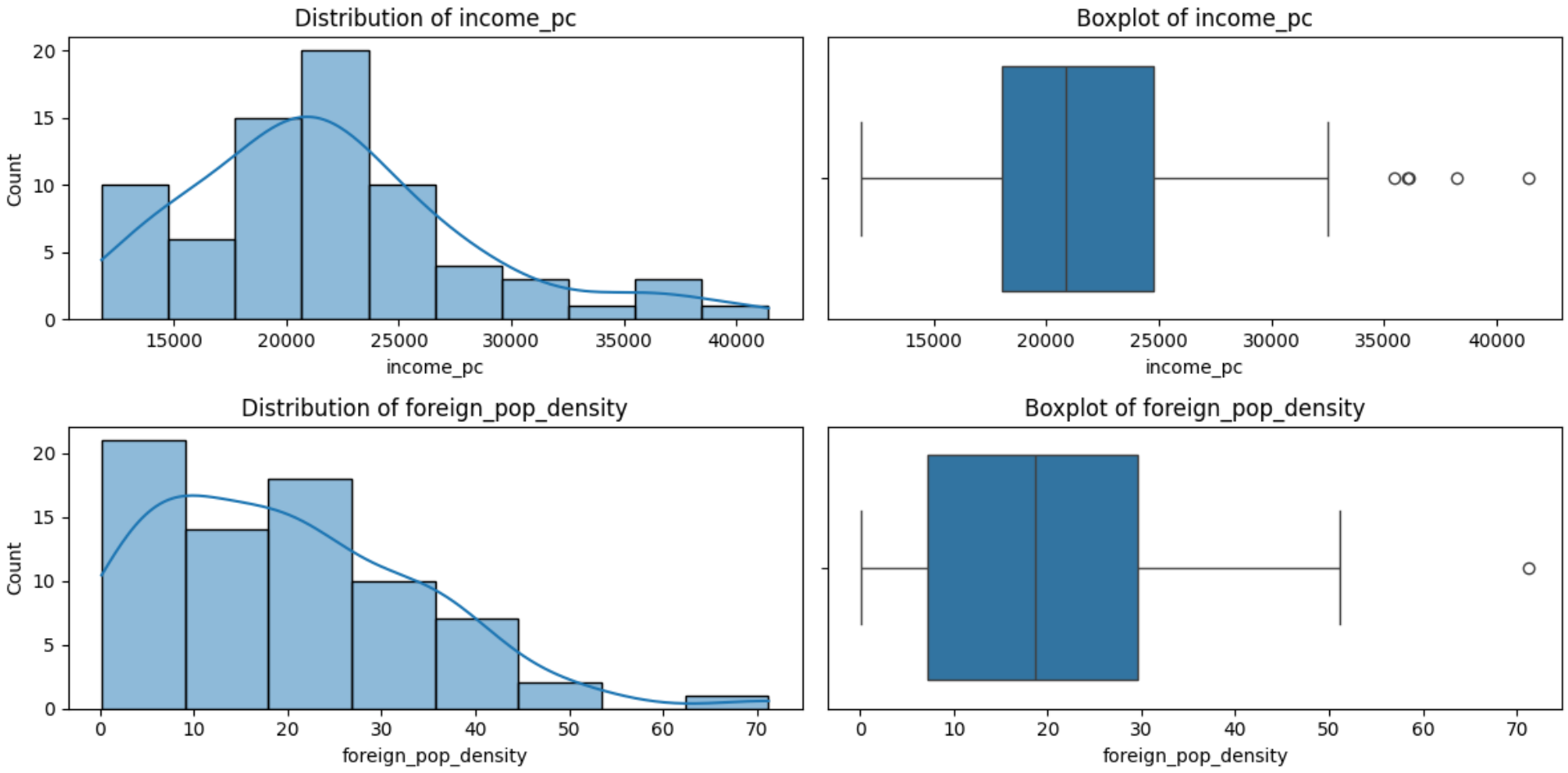
- Strong right-skewness; a few expensive neighborhoods drive dispersion.

VARIABLES DISTRIBUTION – TOURISM PRESSURE



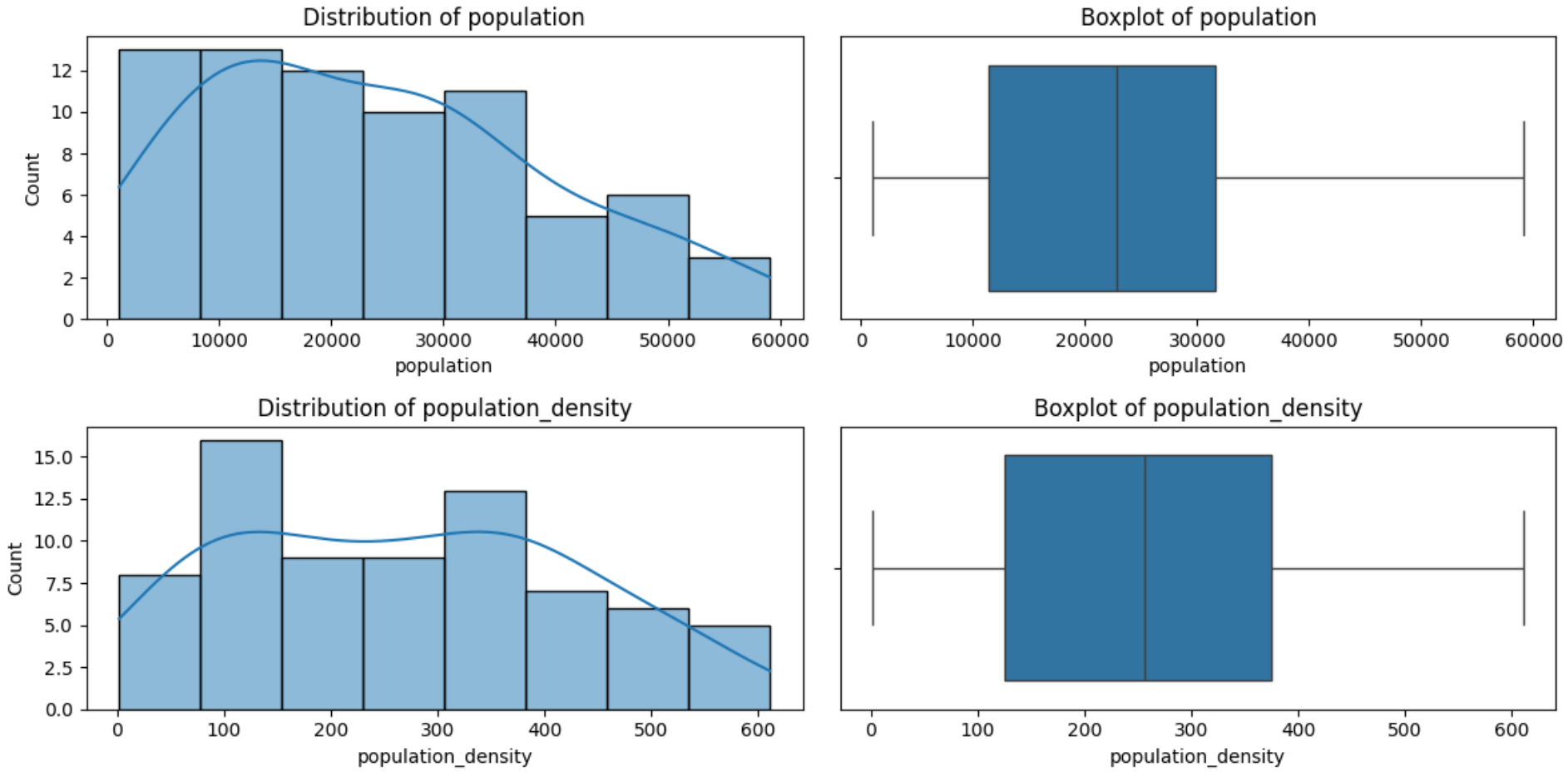
- Tourism pressure is extremely concentrated in central areas.

VARIABLES DISTRIBUTION – SOCIOECONOMIC VARIABLES



- Strong socioeconomic inequality across neighborhoods.

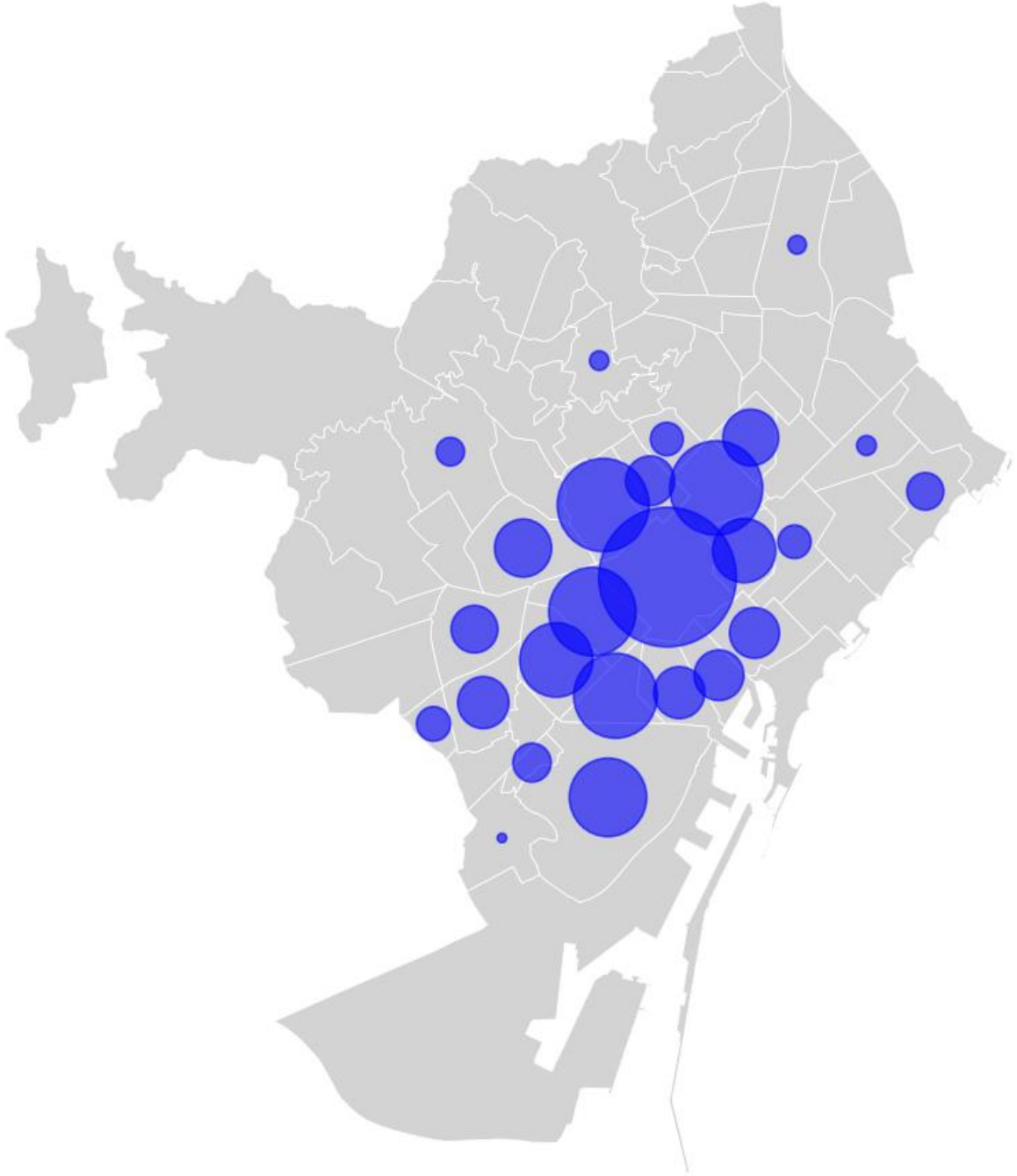
VARIABLES DISTRIBUTION – DEMOGRAFICS



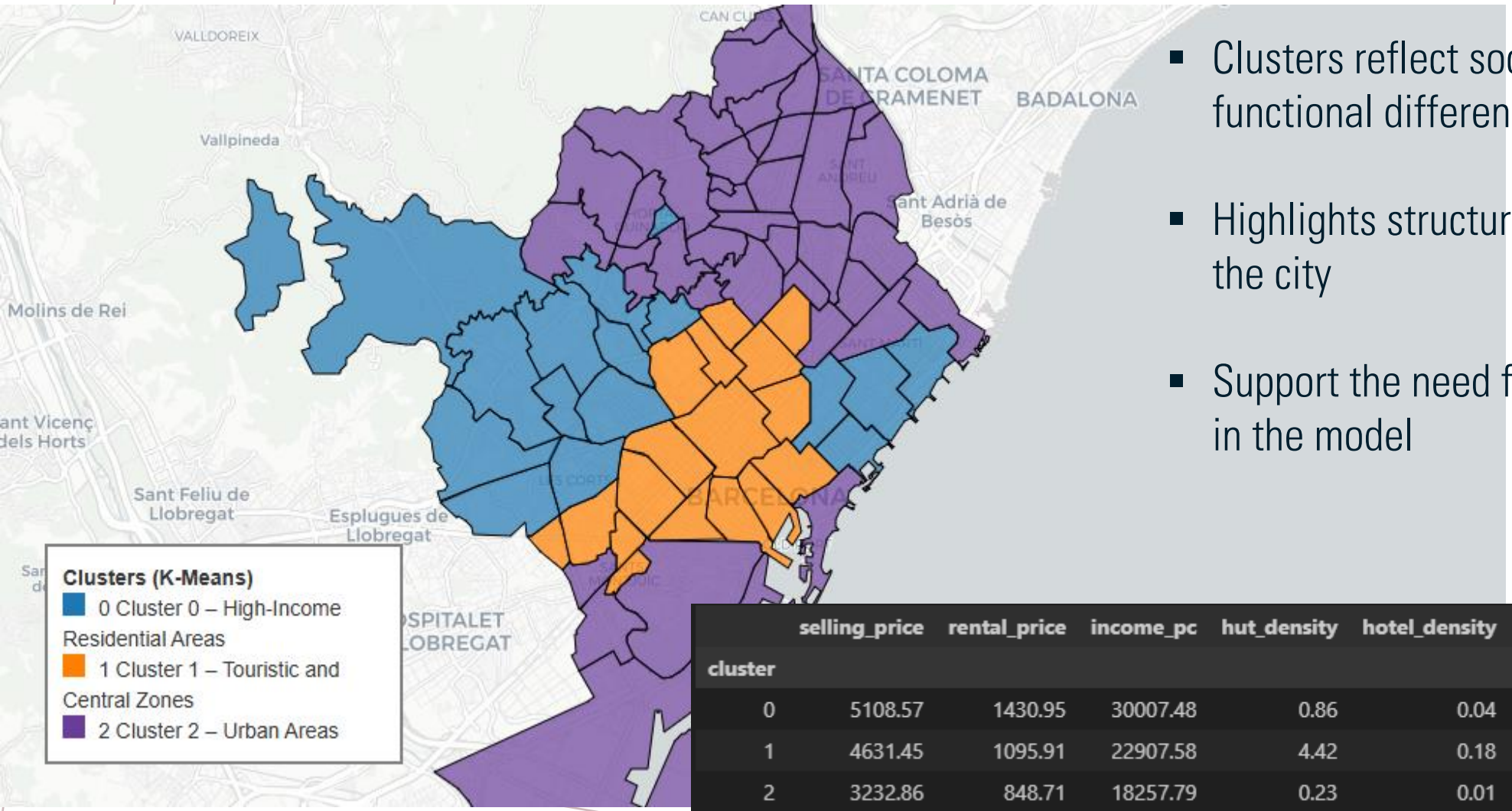
- Demographic densities are extremely uneven, with strong central concentration..

HUTS DENSITY

- Visual concentration of tourist flats
- Clear spatial clusters
- Highlights areas most exposed to policy change



UNSUPERVISED CLUSTERING OF BARCELONA NEIGHBORHOODS (K-MEANS)



- Clusters reflect socio-economic and functional differences
- Highlights structural heterogeneity across the city
- Support the need for econometric controls in the model

cluster	selling_price	rental_price	income_pc	hut_density	hotel_density	foreign_pop_density	population_density
0	5108.57	1430.95	30007.48	0.86	0.04	9.85	161.70
1	4631.45	1095.91	22907.58	4.42	0.18	38.42	423.28
2	3232.86	848.71	18257.79	0.23	0.01	15.60	244.39

MODEL SELLING PRICE

Dependent Variables

- log(selling_price)
- log(rental_price)

Key regressor

- Hut density

Controls

- Income per capita
- Populatin density
- Foreigh Population density
- Hotel density

```
=== OLS: log(Selling Price) ===
                            OLS Regression Results
==============================================================================
Dep. Variable:      log_selling_price   R-squared:                       0.812
Model:                            OLS   Adj. R-squared:                  0.797
Method:                 Least Squares   F-statistic:                     57.70
Date:                Sun, 15 Mar 2026   Prob (F-statistic):           5.98e-23
Time:                        22:53:32   Log-Likelihood:                 46.851
No. Observations:                  73   AIC:                            -81.70
Df Residuals:                      67   BIC:                            -67.96
Df Model:                           5
Covariance Type:            nonrobust
=====================================================================================
                        coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------
const                  -0.3838      0.671     -0.572      0.569      -1.724       0.956
log_hut_density         0.0489      0.040      1.237      0.220      -0.030       0.128
log_income_pc           0.8499      0.070     12.214      0.000       0.711       0.989
log_foreign_pop_density 0.0872      0.066      1.329      0.188      -0.044       0.218
log_hotel_density       0.3146      0.223      1.411      0.163      -0.131       0.760
log_population_density -0.0214      0.051     -0.419      0.676      -0.124       0.081
==============================================================================
Omnibus:                       21.777   Durbin-Watson:                   2.061
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               36.261
Skew:                           1.111   Prob(JB):                     1.34e-08
Kurtosis:                       5.643   Cond. No.                         507.
==============================================================================
```

MODEL RENTAL PRICE

Dependent Variables

- log(selling_price)
- log(rental_price)

Key regressor

- Hut density

Controls

- Income per capita
- Populatin density
- Foreigh Population density
- Hotel density

```
=== OLS: log(Rental Price) ===
                            OLS Regression Results
==============================================================================
Dep. Variable:       log_rental_price   R-squared:                       0.866
Model:                            OLS   Adj. R-squared:                  0.856
Method:                 Least Squares   F-statistic:                     86.47
Date:                Sun, 15 Mar 2026   Prob (F-statistic):           7.47e-28
Time:                        22:53:32   Log-Likelihood:                 59.458
No. Observations:                  73   AIC:                            -106.9
Df Residuals:                      67   BIC:                            -93.17
Df Model:                           5
Covariance Type:            nonrobust
=====================================================================================
                        coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------
const                  -2.5468      0.565     -4.509      0.000      -3.674      -1.419
log_hut_density        -0.0645      0.033     -1.941      0.056      -0.131       0.002
log_income_pc           1.0190      0.059     17.404      0.000       0.902       1.136
log_foreign_pop_density 0.3017      0.055      5.464      0.000       0.191       0.412
log_hotel_density       0.3861      0.188      2.058      0.044       0.012       0.761
log_population_density -0.2839      0.043     -6.597      0.000      -0.370      -0.198
==============================================================================
Omnibus:                       24.834   Durbin-Watson:                   2.175
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               52.423
Skew:                           1.136   Prob(JB):                     4.14e-12
Kurtosis:                       6.475   Cond. No.                         507.
==============================================================================
```

WHY A COUNTERFACTUAL IS NEEDED: INSIGHTS FROM OTHER CITIES

NL Amsterdam

- Partial ban on short-term rentals in 3 central districts (2020).
- Rent prices: little to no measurable impact.
- Sale prices: growth slowed slightly (~2–5%), but no price drop.
- Policy was not city-wide, so not comparable to a full elimination.

ES Palma de Mallorca

- Near-total ban on tourist rentals in apartments (2018).
- Only detached single-family homes allowed.
- Price effects: evidence is mixed, no strong decreases.
- Market is much smaller and more tourism-dependent → not comparable to Barcelona.

COUNTERFACTUAL

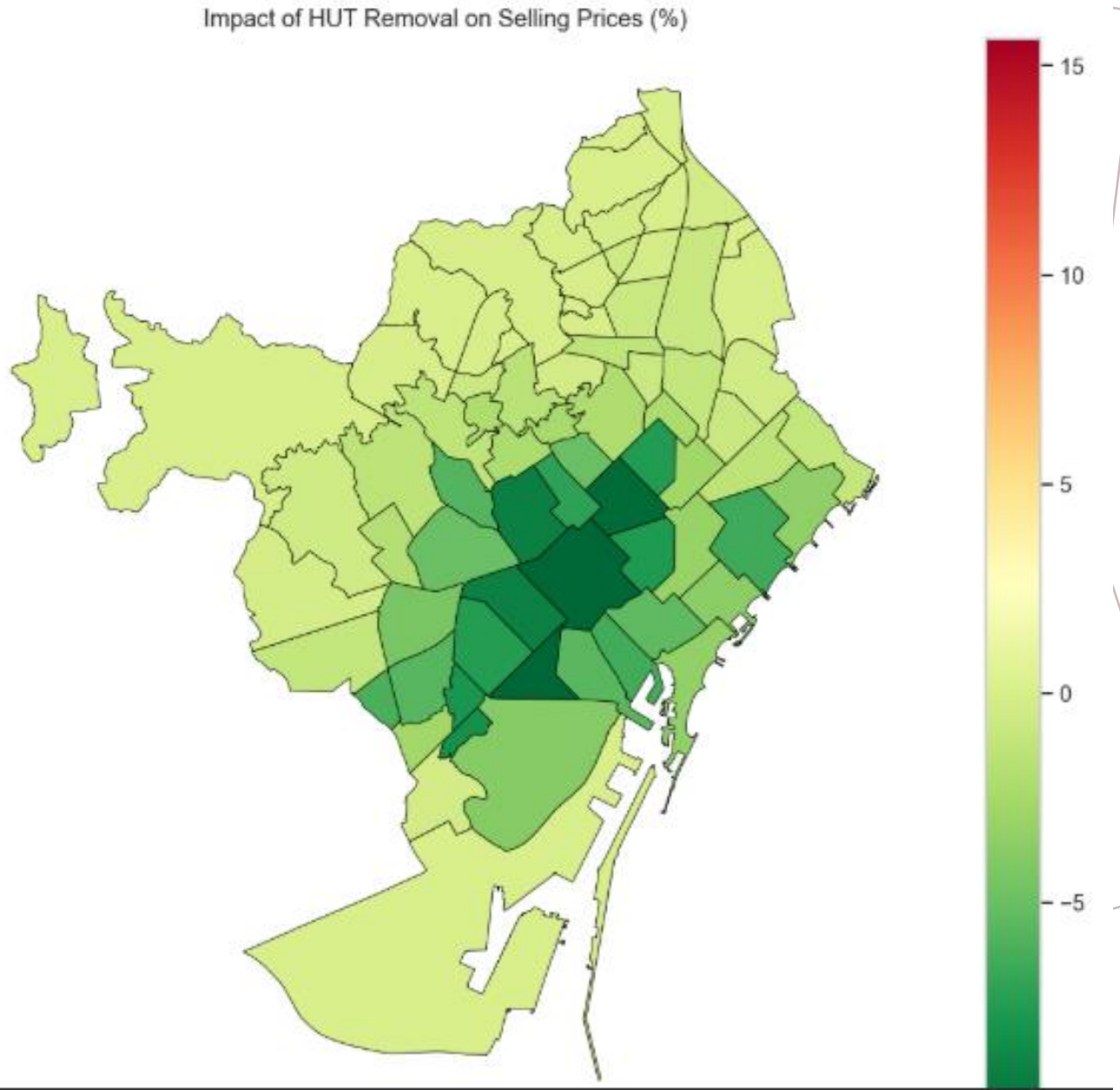
Goal – Simulate the effect of removing all HUTs from the city

- Estimate baseline model
- Predicts prices with actual HUT density
- Set HUT density = 0
- Predict counterfactual prices
- Compute percentage impacts

IMPACT ON SELLING PRICES

- Selling price decrease in high tourism areas
- Strongest effects in central neighborhoods

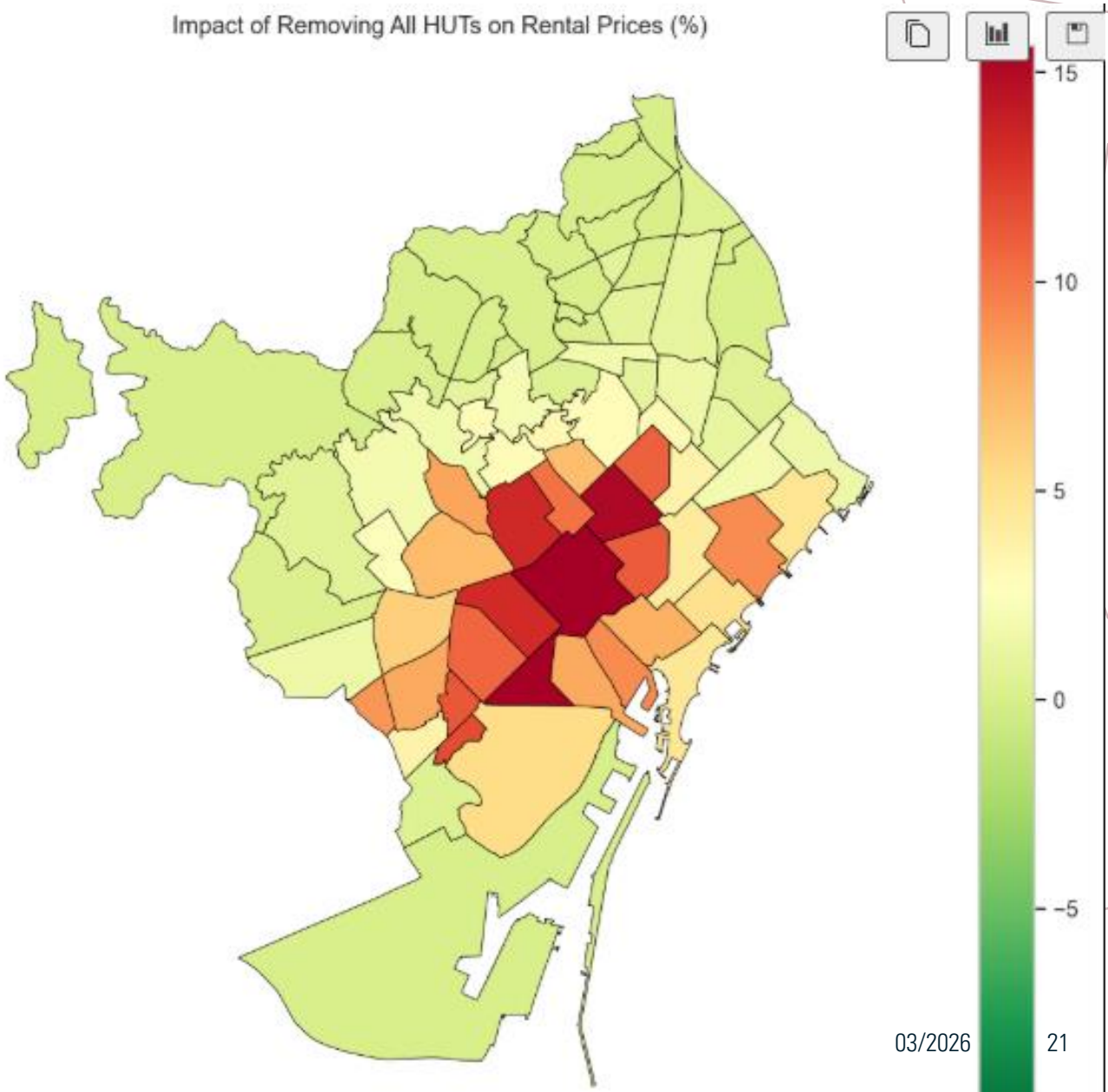
HUT inflate property values



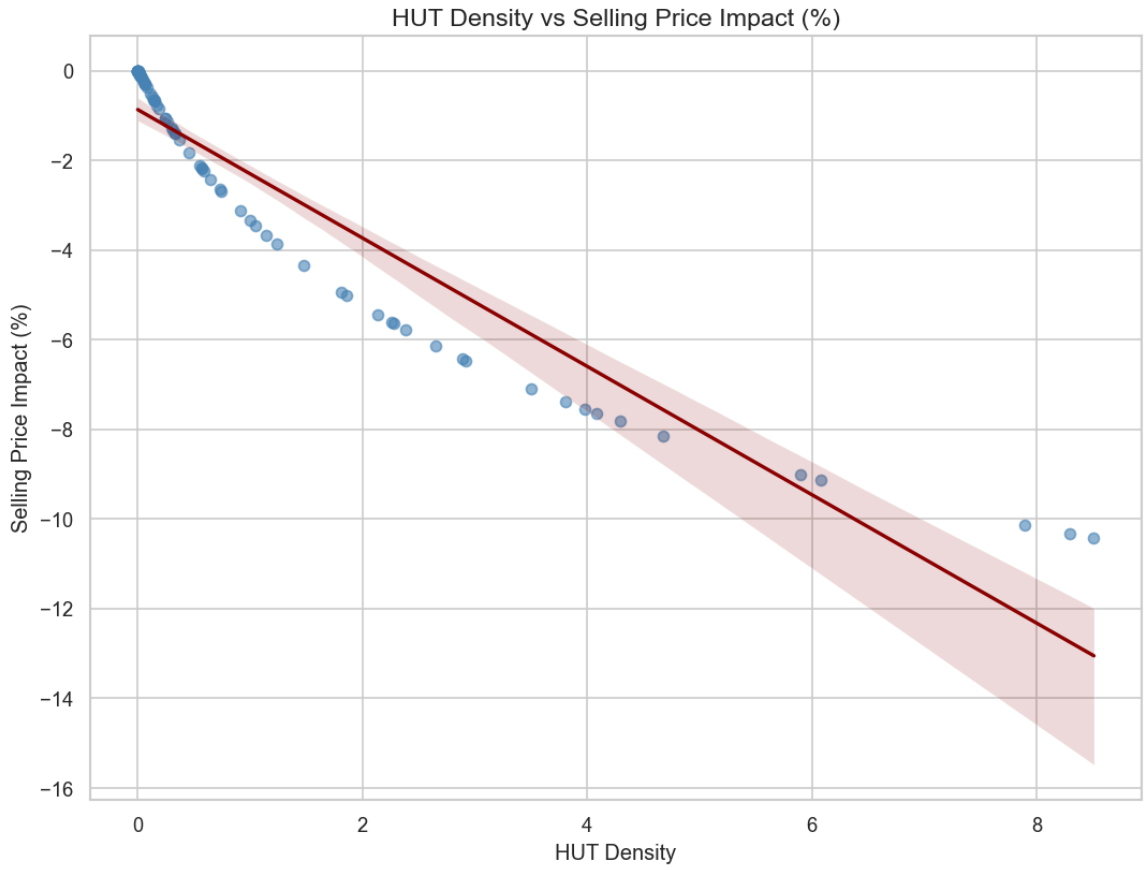
IMPACT ON RENTAL PRICES

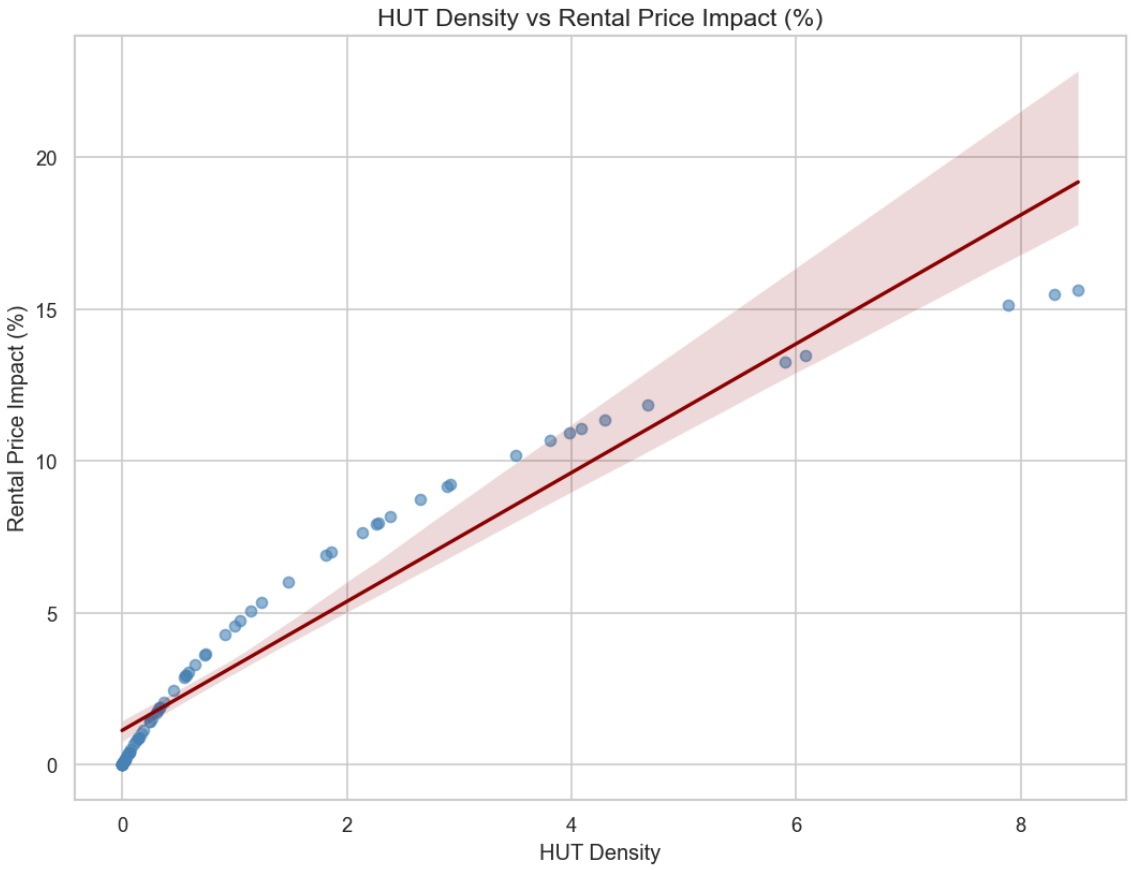
- Rental price increase when HUTs disappear
- Strongest effects in central and dense areas

Demand shifts back to the residential market



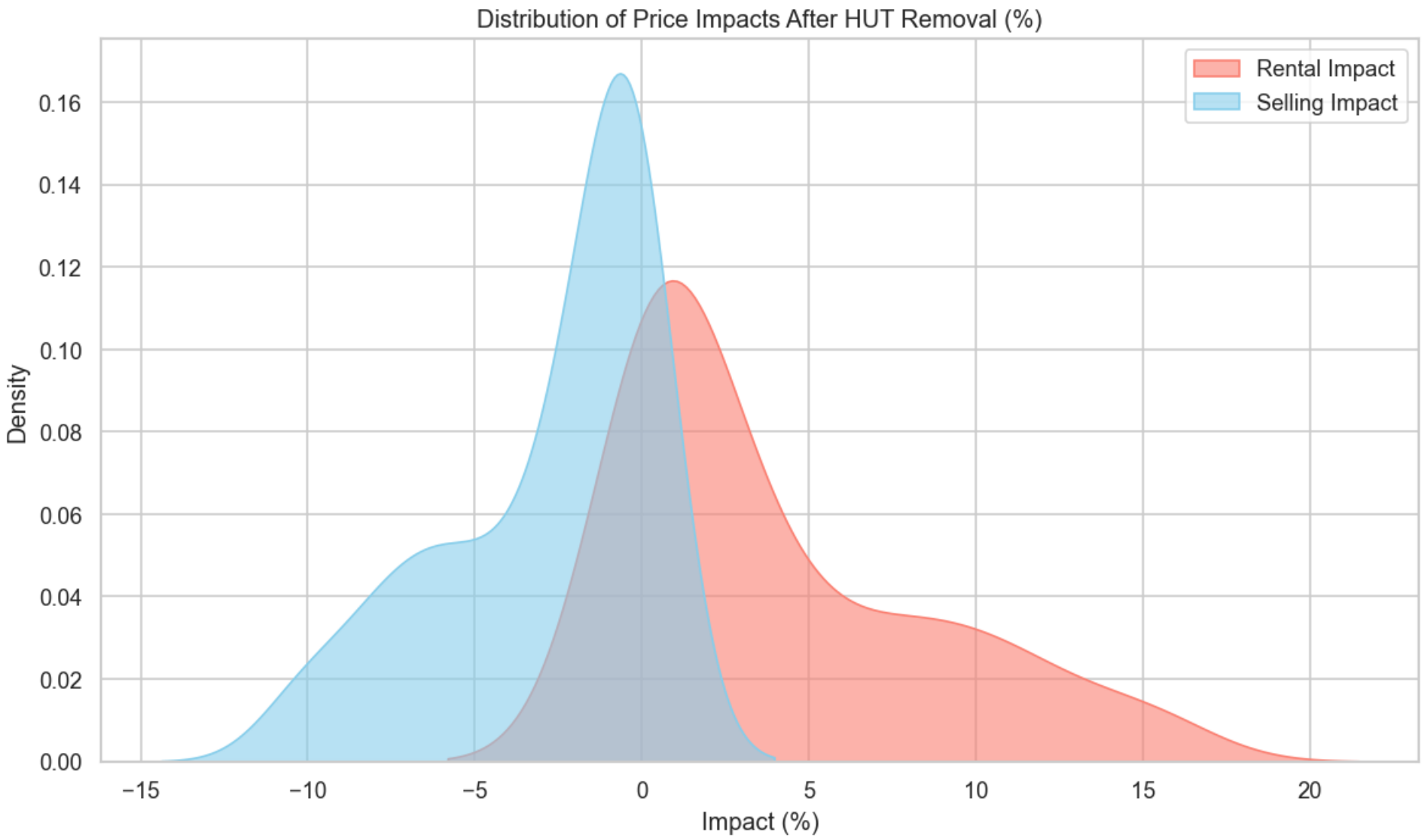
LINEAR RELATIONSHIP





Confirms opposite effect on the two markets

DISTRIBUTION OF IMPACTS



- Selling impacts clusters below zero
- Rental impacts clusters above zero
- Clear assimetry between the two markets
- Strong visual confirmation of the hypotesis

LIMITATIONS

- Model-based scenario, not real-world impact
- No behavioral or long-run adjustments
- Static relationships assumed
- No real control group
- Small rental effect – cautious interpretation

CONCLUSION

Considering the variables in this study:

Removing HUTs

- Lower selling prices
- Raises rental prices

Effects are spatially concentrated

Tourist pressure significantly shapes housing markets

Counterfactual confirms the hypothesis

THANK YOU